

A2 Warm Up 7-5

1. Solve for x , rounding to the *nearest hundredth*.

$$40 \cdot 10^{2x} = 200$$

$$x = 0.35$$

2. A person invests 6500 dollars in a bank. The bank pays 6.5% interest compounded quarterly. To the *nearest tenth of a year*, how long must the person leave the money in the bank until it reaches 11500 dollars?

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

$$8.8 \text{ years}$$

3. Solve for x to the *nearest 10th*.

$$230 = 6000(0.935)^{x+3} - 60$$

$$x = 42.1$$

4. Jaxon invested \$4,500 in an account paying an interest rate of 4.6% compounded annually. Assuming no deposits or withdrawals are made, how long would it take, to the *nearest year*, for the value of the account to reach \$9,750?

$$17$$

5. Solve for a positive value of x .

$$\log_4(256) = x$$

$$4$$

6. Solve the following logarithm problem for the positive solution for x .

$$\log_4 x = -2$$

$$x = 4^{-2} = \frac{1}{16}$$

7. Solve for the exact value of x .

$$\log_2(2x) + 2\log_2(4) = 4$$

$$\frac{1}{2}$$

8. Solve for all values of x .

$$\log_7(3x - 7) + \log_7(2x - 3) = 0$$

$$x = \frac{5}{2}$$

9. Solve for all values of x

$$\log_3(2x^2 + 4x - 7) = 2$$

$$\{x = 2, x = -4\}$$

10. Solve for the *exact* value of x .

$$6 \ln(5x + 4) - 3 = -15$$

$$\frac{e^{-2} - 4}{5}$$